

GAGAN THAKUR

Mandi, Himachal Pradesh

☎ +91 82199 63388 | ✉ talk2gagan09@gmail.com | [linkedin.com/in/gagan-thakur-704779349](https://www.linkedin.com/in/gagan-thakur-704779349) | github.com/seika-afk | [LeetCode](#)

EDUCATION

National Institute of Technology, Hamirpur

Aug. 2024 – Present

B.Tech in Electronics and Communication Engineering

Swami Vivekanand Sr. Sec. School

Apr. 2010 – Apr. 2023

High School

Mandi, India

TECHNICAL SKILLS

Languages:	Python, Rust, JavaScript, C++, C, Assembly
ML / DL Frameworks:	TensorFlow, PyTorch, Keras, scikit-learn, XGBoost
Data Processing:	NumPy, pandas, OpenCV, Matplotlib, Seaborn
LLMs / Generative AI:	LangSmith, LangChain, LangGraph, OpenAI API, RAG
MLOps & Version Control:	Git, GitHub, DVC, MLflow, CI/CD
Cloud & Deployment:	AWS, FastAPI, Docker, Streamlit, Kubernetes, Prometheus

WORK EXPERIENCE

Author & Developer: *Alphabets of Machine Learning*

Apr. 2025 – Present

Self-Driven Open-Source Project

Remote

- Designed and published **26 end-to-end ML projects** spanning supervised, unsupervised, and reinforcement learning with full documentation and reproducible code — used as a learning resource by peers.
- GitHub: github.com/seika-afk/Alphabets-of-Machine-Learning

PROJECTS

Smart Resume Screener [\[GitHub\]](#)

Python, Streamlit, PyMuPDF, SQLite, LLM APIs

- Built an AI-powered resume screening pipeline that parses PDF resumes and extracts structured candidate data (skills, experience, contact info) .
- Evaluated resumes against job descriptions, achieving **75% match accuracy** on a curated test set of 40+ resume-JD pairs.

Vehicle Insurance Cross-Sell Prediction – MLOps Pipeline [\[GitHub\]](#)

Python, DVC, MLflow, FastAPI, Docker

- Built a full MLOps pipeline with DVC for data versioning, MLflow for experiment tracking (**20+ tracked runs**), and a Dockerized FastAPI REST inference endpoint.
- Implemented a CI/CD workflow for automated re-training and deployment on new data snapshots.

NeuroBlock – Deep Learning Library from Scratch [\[GitHub\]](#)

C++

- Engineered a modular C++ deep learning library implementing ANN layers, tensor operations, and backpropagation from scratch without any external ML framework.
- Designed clean layer APIs with serialization and configurable training loops to maximize educational clarity and extensibility.

Conversational Chatbots – 3 Architecture Approaches [\[GitHub\]](#)

Python, TensorFlow, PyTorch, LangChain

- Implemented and compared three chatbot architectures: RNN Encoder-Decoder, a custom Transformer from scratch, and a RAG-based GenAI system using LangChain + OpenAI + vector search — demonstrating progression from classical to modern LLM-based NLP.

ACHIEVEMENTS

- Contributed merged pull requests to major open-source organizations: **Supabase**, **Fossology**, and **SunPy**.
- Solved **LeetCode Top 150** series twice, reinforcing strong foundations in data structures, algorithms, and problem-solving under constraints.

INTERESTS

- Digital art and creative visual design; literature and story writing; personal projects at the intersection of ML and systems programming.